

Do urbanization tasks need both views?

4-model sweep — full vs blind

4-model sweep · full vs blind · 9 urbanization tasks

task	Gemma-4-12B		Qwen2.5-VL-7B		Qwen3.5-4B		Qwen3.5-9B	
	full	Δ bl	full	Δ bl	full	Δ bl	full	Δ bl
Land use	47.3	+29.5	55.6	+1.2	55.6	+17.4	53.2	+14.5
Building height	61.6	+29.0	57.9	+13.2	61.6	+15.3	56.8	+11.0
Urban density	37.3	+14.7	41.8	+16.9	36.3	-11.4	39.0	-10.9
Junction type	66.5	+59.0	57.5	+10.5	50.0	+3.0	55.7	+14.1
Green space	58.6	-17.8	81.8	+3.0	41.4	-12.3	36.9	-24.7
Amenity richness	35.6	+11.4	31.8	+17.5	35.4	+16.4	36.3	+15.2
Road type	65.3	+50.3	50.6	+25.2	57.0	+25.9	53.0	+13.1
Road surface	78.2	+51.8	90.8	+8.0	91.4	+3.9	85.5	+32.7
Transit density	28.0	+1.2	28.0	+1.4	34.2	+7.1	31.4	+5.0

full = both views · Δ bl = full — blind (vision gain) · thinking off · n=190–421/task

Single-model — all four views

task	full	sat	sv	blind	full—best1
Land use	53.0	50.1	52.7	43.5	+0.3
Building height	61.1	47.9	64.7	45.8	-3.6
Urban density	38.5	39.2	44.9	40.4	-6.4
Junction type	62.6	48.8	62.6	48.5	+0.0
Green space	36.5	35.5	37.9	64.5	-1.4
Amenity richness	36.8	36.1	33.5	19.5	+0.7
Road type	47.5	40.1	57.5	34.9	-10.0
Road surface	82.2	78.9	87.1	67.7	-4.9
Transit density	32.1	30.9	30.4	27.3	+1.2
overall	48.6	44.4	50.9	40.9	-2.3

sat = satellite only · sv = street-view only · full—best1 = full — max(sat, sv) · thinking off